

UAMM (Uncertainty-Aware Memory Modulation) — Phase 3

$$\text{Max-Normalization: } \hat{w}_i = \frac{w_i}{\max_j w_j + \varepsilon}$$

best modality = 1.0, others $\in (0, 1]$

Example: daytime scene

Cross-modal weights:
 $w = [0.40, 0.35, 0.25]$
(rgb, thm, lid)

UAMM scores:
 $\hat{w} = [1.00, 0.875, 0.625]$
RGB dominates (best visible)
Others relatively suppressed

Example: night scene

Cross-modal weights:
 $w = [0.20, 0.50, 0.30]$
(rgb, thm, lid)

UAMM scores:
 $\hat{w} = [0.400, 1.000, 0.600]$
Thermal dominates (night vision)
RGB heavily suppressed

P9 UAMM Forward Flow

$$\text{uamm_scores}(B, 3) \hat{w}_i = w_i / \max_j(w_j) \quad \text{best} = 1.0$$

